MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **CBSE CLASS 10 – PERIODIC TEST (SET – B)** | **Time:** 1 Hour
Class: Class X (CBSE) | **Chapter 7: Coordinate Geometry** | **MM: 30**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. This question paper contains 13 compulsory questions divided into three sections: A, B, and C.
2. **Section A** comprises 5 questions of 1 mark each (Q1–Q3 are MCQs; Q4–Q5 are Assertion–Reasoning).
3. **Section B** comprises 5 questions of 2 marks each (Short Answer Type questions).
4. **Section C** comprises 3 questions of 5 marks each (Long Answer Type questions).
5. Use of calculators or electronic devices is strictly prohibited. Draw neat figures wherever required.

SECTION A: 1 Mark Questions (3 MCQs + 2 A–R) (5 × 1 = 5 Marks)

Q1. The distance between the points $(m, -n)$ and $(-m, n)$ is: [1]

(A) $\sqrt{m^2+n^2}$ (B) $m+n$ (C) $2\sqrt{m^2+n^2}$ (D) $\sqrt{2m^2+2n^2}$

Q2. The line segment joining the points $P(-3, 2)$ and $Q(5, 7)$ is divided by the y-axis in the ratio: [1]

(A) 3 : 5 (B) 5 : 3 (C) 2 : 7 (D) 7 : 2

Q3. The coordinates of the midpoint of the line segment joining the points $(-5, 4)$ and $(7, -8)$ are: [1]

(A) $(1, -2)$ (B) $(-1, 2)$ (C) $(2, -4)$ (D) $(6, -6)$

Directions for Q4 & Q5: A statement of Assertion (A) is followed by Reason (R). Choose:
(A) Both (A) and (R) are true and (R) is correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not correct explanation of (A).
(C) (A) is true but (R) is false. **(D)** (A) is false but (R) is true.

Q4. Assertion (A): The distance of the point $(0, 8)$ from the origin is 8 units.
Reason (R): The distance of any point (x, y) from the origin is given by $\sqrt{x^2+y^2}$. [1]

Q5. Assertion (A): If the points $(1, 2)$, $(4, y)$, $(x, 6)$ and $(3, 5)$ are the vertices of a parallelogram taken in order, then $x = 6$ and $y = 3$.
Reason (R): The diagonals of a parallelogram bisect each other. [1]

SECTION B: Short Answer Type Questions (5 × 2 = 10 Marks)

Q6. If the point $P(x, y)$ is equidistant from the points $A(5, 1)$ and $B(-1, 5)$, prove that $3x = 2y$. [2]

Q7. Find the coordinates of the point on the x-axis which is equidistant from the points $(2, -5)$ and $(-2, 9)$. [2]

SECTION B (Continued): Short Answer Type Questions (Questions 8 to 10)

Q8. Find the ratio in which the line segment joining the points $A(3, -3)$ and $B(-2, 7)$ is divided by the x-axis. Also find the coordinates of the point of division. [2]

Q9. If the coordinates of one end of a diameter of a circle are $(2, 3)$ and the coordinates of its centre are $(-2, 5)$, find the coordinates of the other end of the diameter. [2]

Q10. If $A(-2, 1)$, $B(a, 0)$, $C(4, b)$ and $D(1, 2)$ are the vertices of a parallelogram $ABCD$, find the values of a and b. Hence, find the length of side AB. [2]

SECTION C: Long Answer Type Questions ($3 \times 5 = 15$ Marks)

Q11. The vertices of a triangle ABC are $A(3, 0)$, $B(6, 4)$ and $C(-1, 3)$.

(i) Calculate the lengths of sides AB, BC and AC, and prove that ΔABC is a right-angled isosceles triangle. [3]

(ii) Find the coordinates of the fourth vertex D such that $ABCD$ forms a square with diagonal BC. [2]

Q12. A straight line with equation $2x + y - 4 = 0$ intersects the line segment joining the points $A(2, -2)$ and $B(3, 7)$.

(i) Find the ratio in which the line divides the segment AB internally. [3]

(ii) Hence, determine the exact coordinates of the point of intersection. [2]

Q13. The coordinates of the vertices of ΔABC are $A(7, 2)$, $B(9, 10)$ and $C(1, 4)$. If E and F are the midpoints of sides AB and AC respectively:

(i) Find the coordinates of midpoints E and F. [2]

(ii) Calculate the lengths of segments EF and BC using the distance formula. [2]

(iii) Hence, verify and prove that $EF = \frac{1}{2}BC$. [1]

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